



Disha AI – AI-Based Disaster Prediction, Detection and Response System

AI-Powered Smart Disaster Management

PRESENTATION 2026

TEAM

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DishaAI

AI-Powered Disaster Prediction,
Detection and Response System

Guide
Rahul Nawkhare

PROJECT OUTLINE

- **Understanding the Problem** – Identify the major challenges in disaster prediction, detection and emergency management.
- **AI-Based Solution** – Introduce DishaAI and how AI technologies can improve disaster management.
- **System Design** – Explain the architecture, technologies and major components of the system.
- **System Working** – Demonstrate how DishaAI collects, processes and analyses disaster-related data.
- **Intelligent Response** – Show how the system generates alerts, recommendations and emergency assistance.
- **Results & Future Scope** – Discuss the expected benefits, conclusion and possible future improvements.

Problem Statement

Disasters are unpredictable, but their impact is preventable with the right information and response.



Disasters such as floods, earthquakes, cyclones, landslides and fires can cause severe damage to life, property and the environment.



Delayed Prediction and Detection

Existing systems often rely on manual monitoring and face delays in identifying and predicting disasters.



Fragmented Information

Disaster-related data is collected from multiple sources but is not always available in a single platform.



Slow Response

Lack of real-time information leads to slow emergency response and poor decision-making.



Limited Public Awareness

People often struggle to get quick and reliable safety instructions during emergencies.



Lack of End-to-End Solution

Most existing systems focus on only one aspect (prediction, monitoring or response) instead of providing a complete solution.



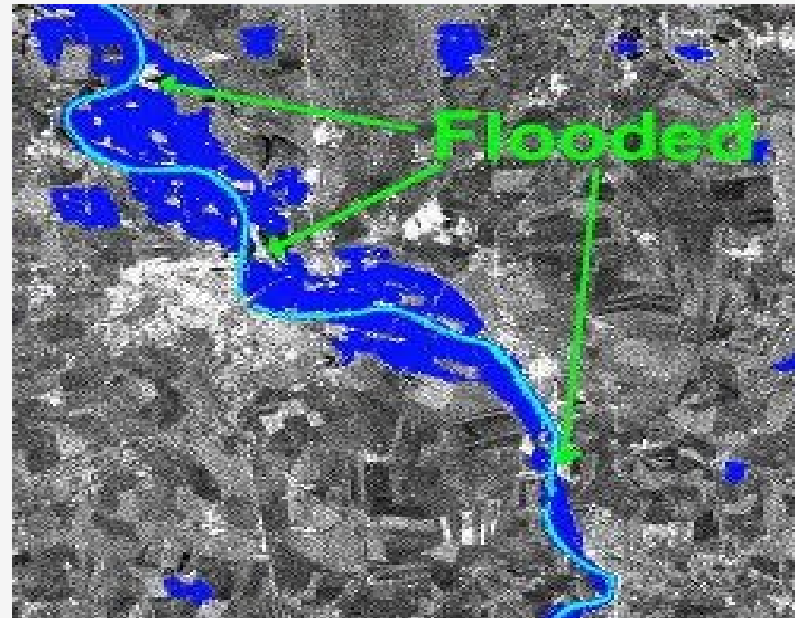
Therefore, there is a need for an intelligent, automated and centralized system that can predict risks, detect disasters and assist with rapid response.



Problem Statement

- Natural and man-made disasters such as **floods, earthquakes, cyclones and fires** can cause severe loss of life and property.
- Existing disaster management systems may face challenges in **early prediction, real-time detection and rapid response**.
- Disaster-related information is often scattered across different sources, making quick decision-making difficult.
- There is a need for an **intelligent, automated and centralized system** that can assist during different stages of a disaster.

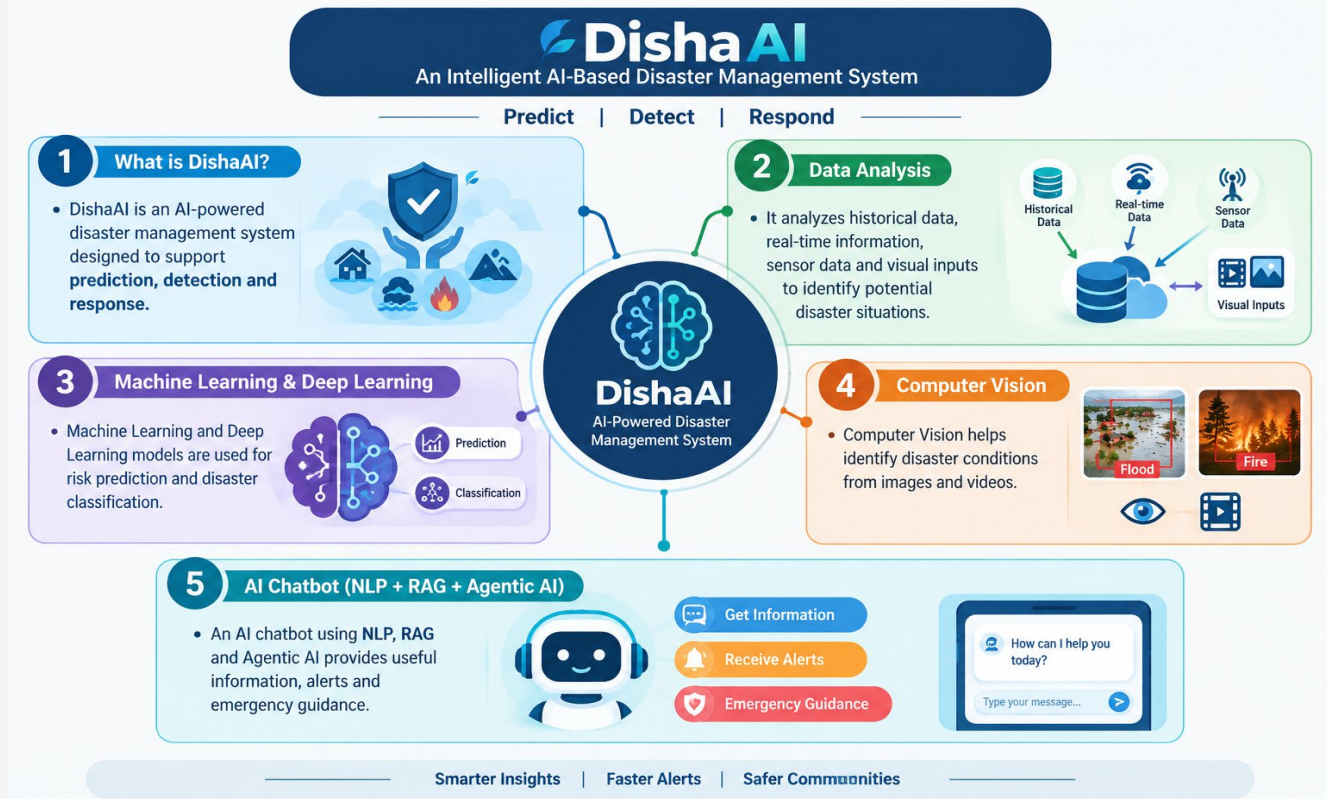




- Natural disasters are extreme events caused by natural processes that can seriously affect human lives, property and the environment.
- Common examples include floods, earthquakes, cyclones, landslides, wildfires and droughts.
- Early prediction, timely detection and rapid response can help reduce damage and save lives

Proposed Solution

- **DishaAI** is an AI-powered disaster management system designed to support **prediction, detection and response**.
- It analyzes **historical data, real-time information, sensor data and visual inputs** to identify potential disaster situations.
- Machine Learning and Deep Learning models are used for **risk prediction and disaster classification**.
- Computer Vision helps identify disaster conditions from **images and videos**.
- An AI chatbot using **NLP, RAG and Agentic AI** provides useful information, alerts and emergency guidance.



Run Local Server:

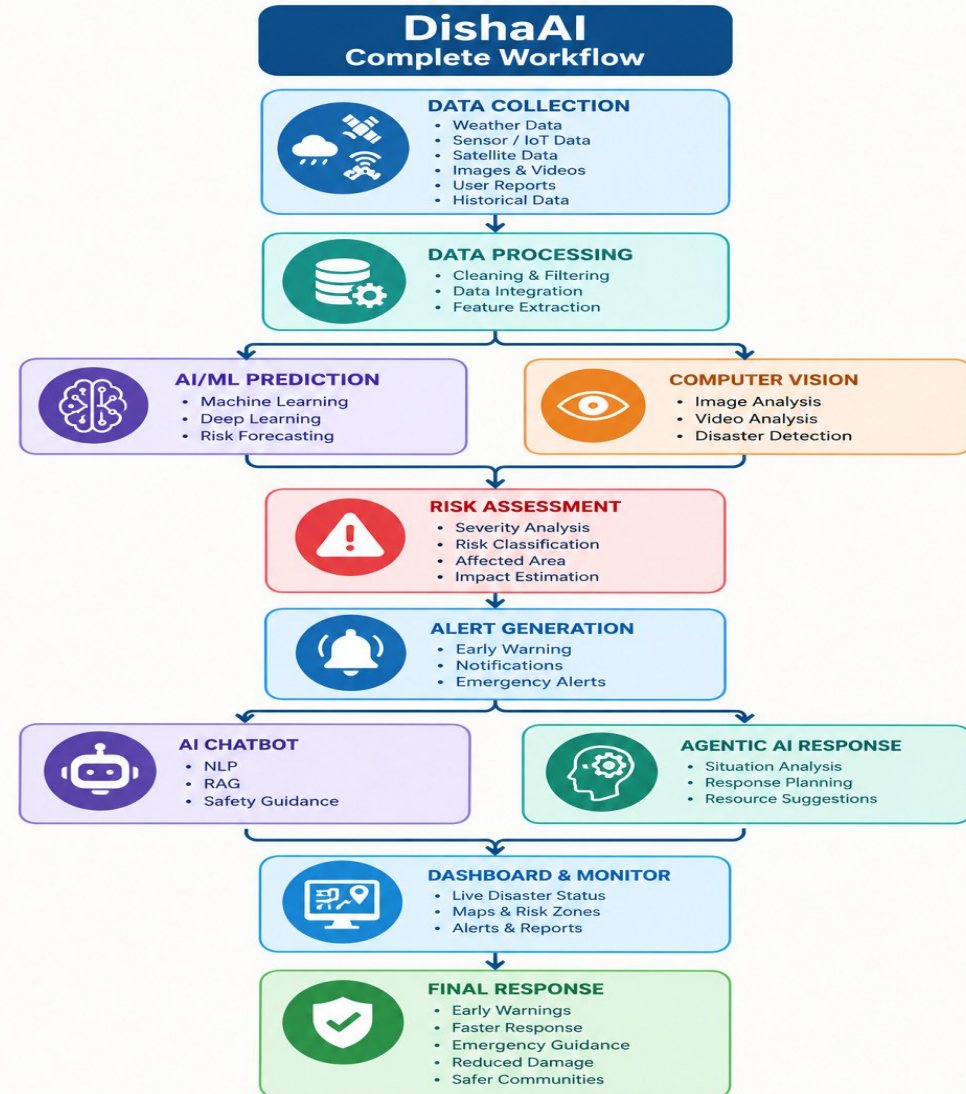
py -m http.server 8080

Open Disha AI:

<http://localhost:8080>

System Architecture

- **Data Collection Layer** – Collects information from sensors, weather sources, images, videos and user reports.
- **Data Processing Layer** – Cleans, processes and prepares the collected data for AI analysis.
- **AI/ML Layer** – Predicts disaster risks and analyses the severity of the situation.
- **Computer Vision Layer** – Detects visible disaster conditions from images and video.
- **NLP, RAG & Agentic AI Layer** – Understands user queries, retrieves relevant information and supports intelligent actions.
- **Response Layer** – Generates alerts, recommendations and emergency instructions.
- **User Interface** – Displays predictions, alerts, reports and chatbot responses through a dashboard.



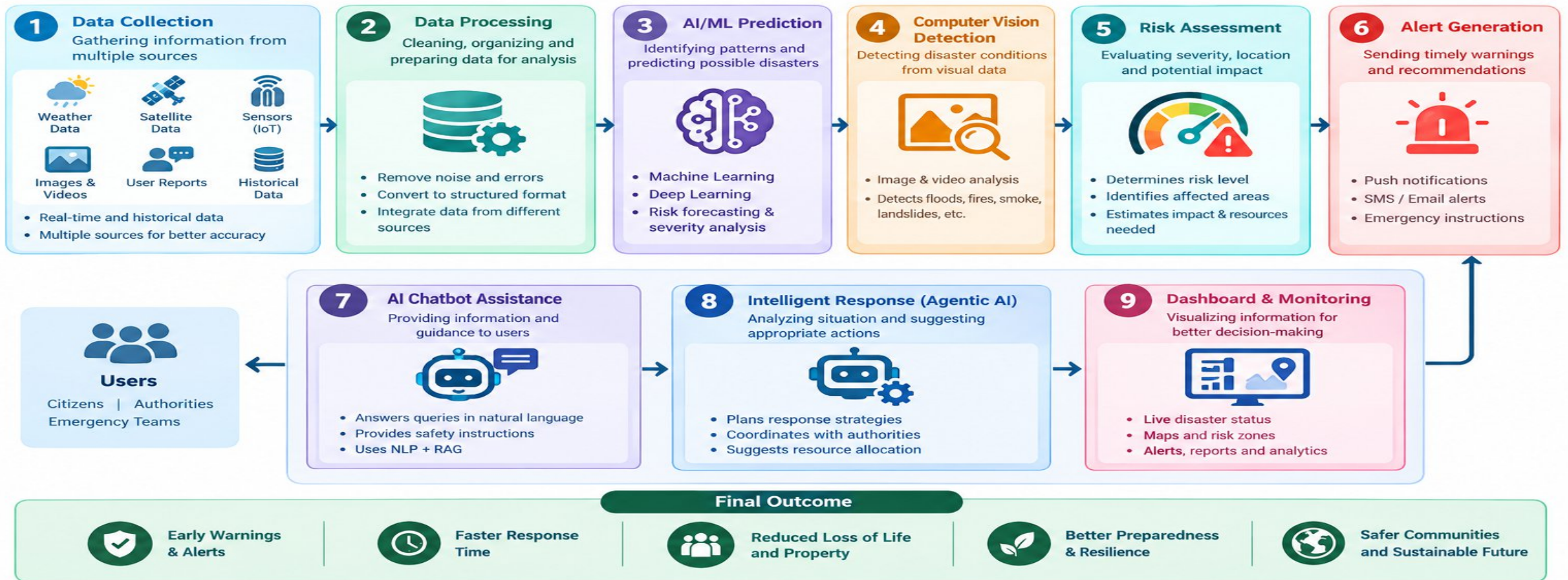
Technologies Used

- **Python** – Used for AI models, data processing and backend development.
- **Machine Learning & Deep Learning** – Used for disaster prediction, classification and risk analysis.
- **Computer Vision** – Used for detecting disaster-related conditions from images and videos.
- **NLP & RAG** – Used to understand user queries and provide accurate disaster-related information.
- **Agentic AI** – Helps automate decision-making and coordinate response actions.
- **React.js** – Used to build an interactive and user-friendly dashboard.
- **FastAPI/Flask** – Used for developing backend APIs and connecting AI models with the frontend.
- **Database & Cloud Deployment** – Used for storing information and making the system accessible in real time.

Working of DishaAI

DishaAI Working Process

From Data to Action – Predict | Detect | Respond



LIVE DEMO OF THE PROJECT

← ↻ ⓘ localhost:8000

⋮ ☆ ⋮ ⋮ ⋮ Chat

LIVE SENTINEL

National Warning Grid: Cyclone Warning #04 active for Bay Coast. 42 elevated relief shelters open. Real-time radar scan updating every 30s.

UTC+05:30 • Live Monitoring: 19:08:42

Sound: Off

**DishaAI**

BE CAPSTONE

Intelligent Disaster Prediction, Detection & Response System

Briefing & Concept Map

Dashboard

20 Disasters Directory

Research & Viva

Layman View

CSE Tech View

Emergency Mode

Sign in with Google

?

⚙

• BE Final Year Capstone Project • AI-Powered Disaster Intelligence

DishaAI: Intelligent Disaster Prediction, Detection & Response System

When disasters strike, confusing scientific alerts cost human lives. **DishaAI** bridges advanced Artificial Intelligence (Machine Learning, Deep Learning, Computer Vision, Multilingual NLP, RAG, and Autonomous Agents) into crystal-clear directions that every everyday citizen can understand.

DUAL EXPERIENCE PLATFORM

Switch anytime between **Layman Citizen View** (simple analogies, safe shelters, checklists) and **CSE Technical View** (neural models, datasets, loss graphs, multi-agent pipelines).

Start with Google Sign-In →

Live Multi-Modal AI Sentinel

Sentinel-1 SAR + River Hydro Telemetry

ONLINE



Cyclone Blip 82%

Your Location

Shelter #1 Open

SAR SATELLITES

Sentinel-1/2

DRONE INFERENCE

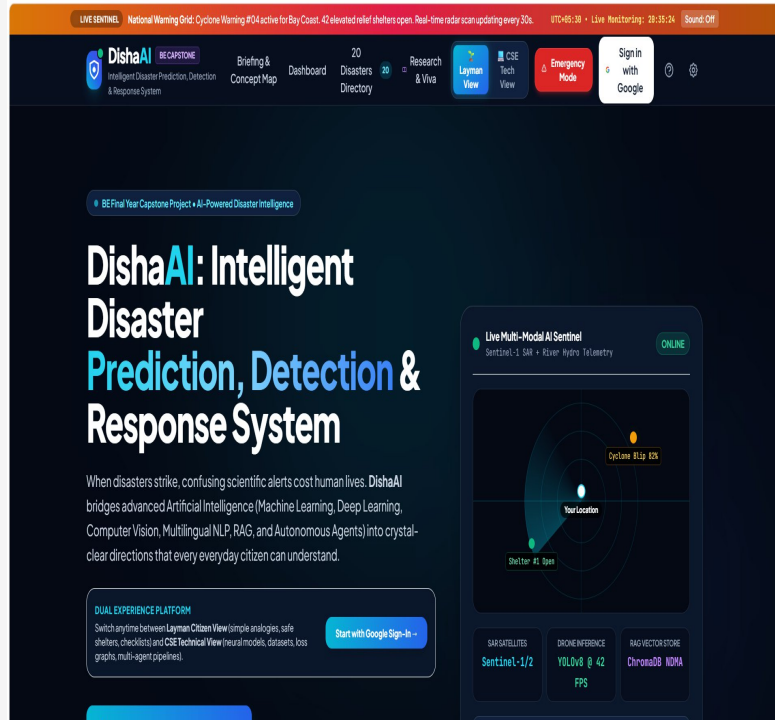
YOLOv8 @ 42 FPS

RAG VECTOR STORE

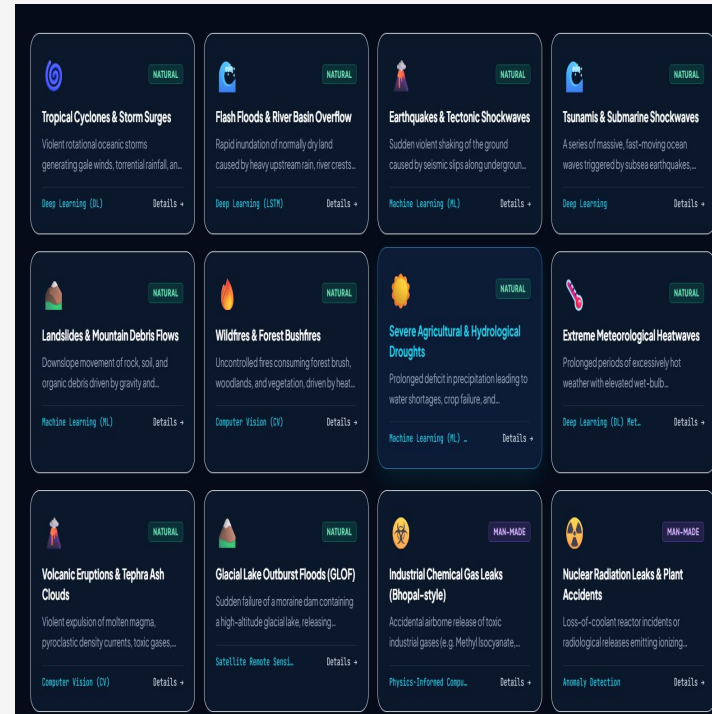
ChromaDB NDMA

🔍 Briefing: What Does DishaAI Do?

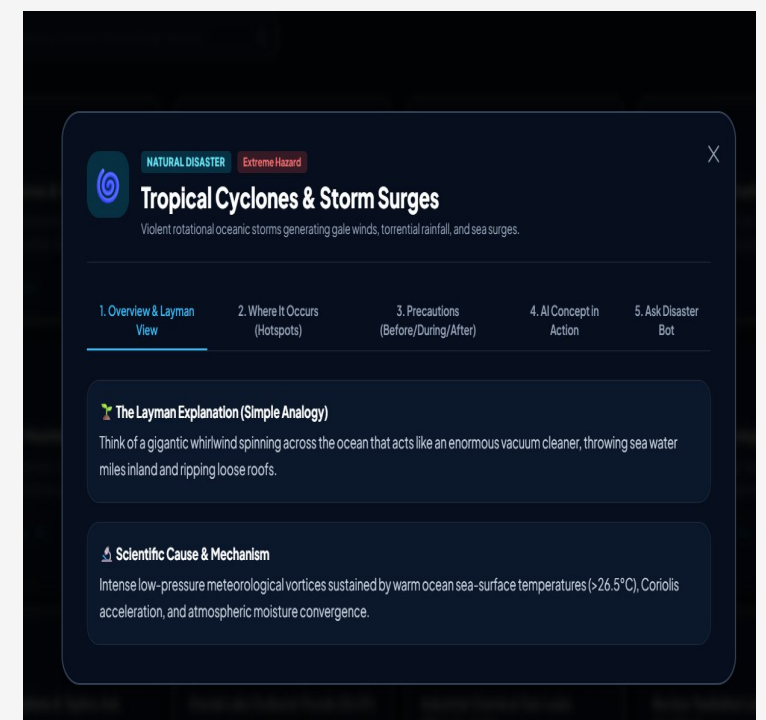
Modules



DishaAI Home/Dashboard Page



List of Natural/Man Made Disasters



Description of each disaster

Conclusion & Future Scope

Conclusion

- **DishaAI** combines multiple AI technologies to create a smart and centralized disaster management solution.
- It helps in early prediction, real-time detection, risk assessment and emergency response.
- The system can improve situational awareness and support faster decision-making during disasters.

Future Scope

- Integration with IoT sensors, satellites and live weather data for better real-time monitoring.
- Development of a mobile application for faster public access to alerts.
- Support for multiple regional languages to reach more users.
- Integration with government agencies, emergency services and rescue teams.
- Development of more advanced autonomous AI agents for real-time disaster response and resource coordination.

THANK YOU